

JadeSpectrogram2 Manual

J. Bitzer, License: CC-BY 4.0

September 29, 2026, for JadeSpectrogram2 1.3.1 (manual build 4)

1 Introduction

Dear User,

a spectrogram is not the sexiest of all plug-ins, but sometimes very versatile. This incarnation comes as a plug-in and as a standalone application and is highly flexible in terms of its analysis parameters. You can tune it for your purposes.

JadeSpectrogram2 is the new version of the JadeSpectrogram, with a new code base. It only analyses: the audio passes through unchanged and without any latency, so you can put it anywhere in your signal chain.

I hope you will enjoy using it and gaining knowledge for acoustical or musical problems.

Have fun!

Joerg aka audio-dsp.

2 Installation

The downloaded file is a standard zip file. It contains a plug-in directory (.vst3), an AU file (.component, macOS only), the Standalone application, a ReadMeFirst.txt, the licence files and this manual. Copy the plug-in to the right place for VST3 / AU plug-ins and let your DAW rescan its plug-ins.

2.1 Windows

For Windows™ the standard VST3-directory is:

C:\Program Files\Common Files\VST3

2.2 Mac VST

If you use the VST3 plugin, copy the .vst3 directory to

/Users/yourUSERNAME/Library/Audio/Plug-Ins/VST3

2.3 Mac AudioUnit

If you use the AudioUnit plugin, copy the .component file/directory to

`/Users/yourUSERNAME/Library/Audio/Plug-Ins/Components`

The macOS version runs natively on Apple Silicon and Intel Macs. It is not signed with an Apple Developer ID yet. If macOS reports that the plug-in is damaged or cannot be opened, remove the quarantine flag in the Terminal, e.g.

```
xattr -cr ~/Library/Audio/Plug-Ins/VST3/JadeSpectrogram2.vst3
```

2.4 Linux

For Linux copy the full directory to:

`/home/yourUSERNAME/.vst3/`

2.5 Standalone

The Standalone version runs without a DAW: it analyses the audio input of your sound card. Choose your audio device and input channels in its settings.

2.6 (De)-Installation

At the first start, the plug-in creates a (still empty) folder for presets. If you remove the plug-in, you can delete this folder as well:

Windows:

`C:\Users\yourUSERNAME\AppData\Roaming\Jade_Hochschule\JadeSpectrogram2`

Mac:

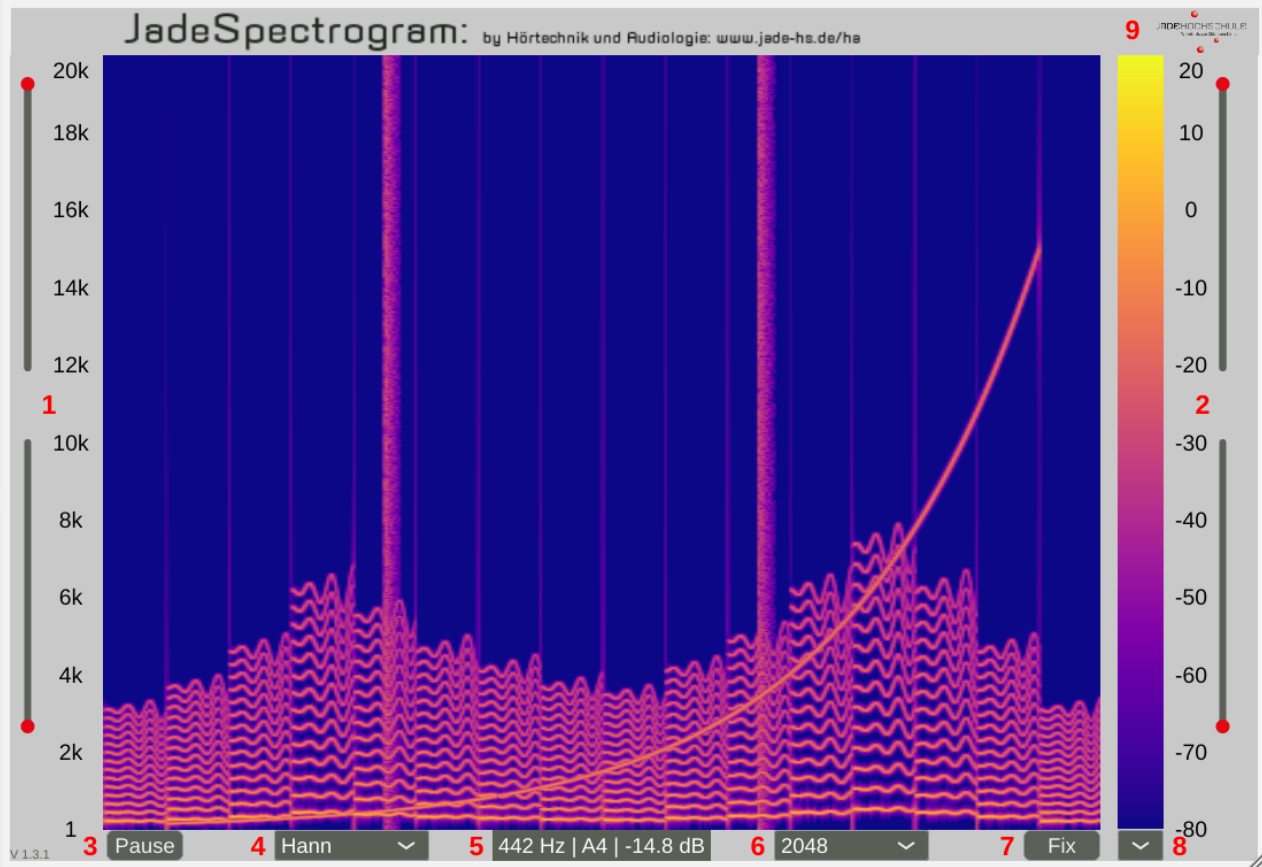
`/Users/yourUSERNAME/Library/Audio/Presets/Jade_Hochschule/JadeSpectrogram2`

Linux:

`/home/yourUSERNAME/.config/Jade_Hochschule/JadeSpectrogram2`

3 How to use

JadeSpectrogram2 has two main areas: the analysis display in the middle, and all parameter settings at the left and right side and at the bottom.



The example shows a short melody with its harmonics (with vibrato), an exponential sine sweep and two noise bursts. The mouse points to the second harmonic of the last note (5).

3.1 Parameters Left (1)

On the left side the two sliders limit the displayed frequency region (lower limit 1 Hz to 10 kHz, upper limit 500 Hz to 20 kHz, at most half the sampling rate). This does not change the analysis; the display shows an interpolated version of the original data. A small frequency interval between upper and lower frequency leads to blurry content.

3.2 Parameters Right (2)

On the right side the two sliders change the upper and lower boundaries of the colourmap (between -80 and $+20$ dB). This allows for higher contrast in the level region you are most interested in. The colour bar next to them shows the colourmap.

3.3 Parameters Bottom (3,4,5,6,7,8)

- Pause button (3): stops the analysis, the display freezes for a closer look. The audio still passes through. Click again to continue.

- Window selector (4): the short-time Fourier transform divides the signal into short blocks and weights each block with a window. Different windows trade frequency resolution against leakage: Rectangular, Hann, Hamming, Blackman-Harris, Flat Top and Hann-Poisson. When in doubt use the Hann window as a starting point; Flat Top is best for reading exact levels of sinusoids, Blackman-Harris shows weak components next to strong ones.
- Readout (5): this small display shows the frequency in Hz, the note name ($A4 = 440$ Hz) and the level in dB at the current mouse position in the analysis display. This is a very versatile tool, e.g. to analyse melody tracks or to read the level of a harmonic.
- FFT size (6): changes the block size and the FFT size of the analysis (512 to 8192). Use higher values for detailed frequency analysis and lower values for timing analysis, see table 1. Changing the size restarts the display.
- Display mode (7): by default the spectrogram scrolls from right to left, the newest data is at the right edge. Click **Fix** to freeze the image in place; a red line then runs from left to right and overwrites the oldest data. Click **Run** to scroll again. The button always shows what a click will do.
- Colourmap selector (8): change the colourmap to your taste (Mono, BW, Hot, Rainbow, Viridis, Plasma, Jade; default: Plasma). If you want to show results to others keep in mind that some colourmaps are better suited than others. Modern colourmaps (viridis or plasma) are equally spaced in their brightness perception and suitable for colourblind people.

FFT size	block length	frequency spacing	new column every
512	10.7 ms	93.8 Hz	5.3 ms
1024	21.3 ms	46.9 Hz	10.7 ms
2048	42.7 ms	23.4 Hz	21.3 ms
4096	85.3 ms	11.7 Hz	42.7 ms
8192	170.7 ms	5.9 Hz	85.3 ms

Table 1: Time and frequency resolution at a sampling rate of 48 kHz. The blocks overlap by 50 %.

3.4 About box (9)

Click on the Jade Hochschule logo to show the about box, click anywhere to close it.

3.5 Analysis Display

The huge main area is the analysis display. It shows the last 8 seconds of the analysed signal on the x-axis vs frequency on the y-axis. The power of the frequency content is represented by different colours. Usually, the brighter the colours, the more power there is at that frequency. For a stereo signal, the power spectra of both channels are averaged.

The level values are a power spectral density in dB relative to digital full scale. They are not calibrated with regard to sound pressure level (SPL), and the absolute values depend on the window and the FFT size. Use them to compare levels within the same setting.

3.6 What is stored with your project

Your DAW project stores the four slider settings (frequency range and colour range) and the size of the plug-in window (drag the lower right corner to scale it). FFT size, window, colourmap and display mode are not stored yet.

4 Further topics

If you want to dive into signal analysis, the spectrogram is just the beginning. I really recommend the web-site of Prof. Julius O. Smith III <https://ccrma.stanford.edu/~jos/sasp/>.

4.1 Source code

This plug-in is open source. You can find the source code here: (<https://github.com/JoergBitzer/JadeSpectrogram2>)

Everything needed to build it is in this repository (clone it with `-recursive` to get JUCE and the TGMStaticLib as submodules, then build it with CMake). The binaries for Windows, macOS and Linux are built automatically by GitHub Actions.

This plug-in belongs to a longer list of plug-ins developed for educational purposes. If you are familiar with JUCE, you will have no problem to understand what's going on. Please report issues at the GitHub repository.

4.2 More Web links

<https://www.musicdsp.org/>

5 Legal stuff

For this documentation the CC-BY 4.0 licence is valid. Copyright holder is J. Bitzer. The source code of the plug-in is under the MIT licence. The plug-in binaries contain JUCE (used under the AGPLv3), the VST3 SDK (MIT), on macOS the Audio Unit SDK (Apache 2.0), the FFT by Uwe Simmer (MIT-style licence) and colourmap data from matplotlib (CC0); the binaries as a whole are distributed under the AGPLv3. The licence texts are included in the download.

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- Mac[®] is a trademark of Apple Inc.
- VST[®] is a registered trademark of Steinberg Media Technologies GmbH

A Release notes

1.3.1 (September 2026)

- Zero latency: the audio passes through unchanged. Before, it was delayed by half the FFT size (up to 4096 samples), and the delay was not reported to the host.
- Changing the FFT size or the window while the audio is running no longer causes dropouts: the analysis switches without locks and without memory allocation in the audio thread.
- Much lower memory use (about 0.5 GB less).
- The FFT size is kept when the audio device restarts, and the FFT size selector shows the right value when you reopen the plug-in window.
- Default colourmap is now Plasma (before, the selector and the display did not agree).
- Bug fixes: possible crash of the readout at the top of the display (sampling rates up to 40 kHz); the label of the display mode button (7) did not change at the first click.
- Downloads for Windows, macOS (Universal: Apple Silicon and Intel) and Linux, built automatically from the source code.
- The changes of this release were developed with the help of an AI (Anthropic's Claude, in Claude Code) and reviewed by J. Bitzer.

1.1 – 1.2.1 (February 2026): JadeSpectrogram2

- New, self-contained code base: everything needed to build is in one repository, no dependency on Eigen anymore; the analysis uses the fast internal FFT.
- The data exchange between audio and display is block free (FIFO).
- The readout (5) also shows the level in dB.
- More exact scaling.
- Bug fixes: the frequency sliders cannot overlap anymore; negative mouse positions in the readout.

1.0 (2021): JadeSpectrogram

- First release (VST3 and AU), based on the TGM development framework.